

Manual

Energy Management Consumption Correlation EMG-KLE 8.3

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1 Energy Management – Consumption Correlation

Purpose

Evaluation function for energy data. The function provides efficient graphic evaluations that you can use to monitor the development of energy consumption (load profile) in correlation with produced orders and articles or the machine status.

You use the function package for the following purposes:

- You want to analyze consumption over time. Here, the time is partitioned in small intervals of 15 minutes each.
- You want to analyze consumption in relation to the produced articles or orders.
- You want to analyze consumption in relation to the machine status.

Integration

The evaluations require collected data and are therefore based on the function packages EMG-MGM and EMG-EVF.

Features

- Graphic presentation of energy consumption over time. Display using intervals of 15 minutes or hour intervals. Display of load profile as trend line.
- Hierarchical selection of the displayed energy counters. Display of calculated or physical counters.
- Graphic GANTT chart showing the operations logged on during the relevant period, narrowed down by workplace, order number or article.
- Graphic GANTT chart showing the machine statuses posted during the relevant period, narrowed down by workplace or (report) group.
- Projection of consumption data to correlated bars in the chart using a multiple selection of operations/machine statuses including a totals function for the total time.

2 Consumption correlation

Overview

Menu	Resource management → Resource analysis → Consumption correlation
Transaction code	concor
Function authorization	concor

This document describes the "consumption correlation" application in the Manufacturing Operation Center (MOC).

Application

The consumption correlation graphically shows energy consumption in relation to the recorded operations. You can present energy consumption in connection with the accrued information on operations.

Energy documents are assigned to the produced operations by time and selection.

Selection criteria

Reference

Selection of resources using the resource list

Date from / until

Selection of a period for consumption correlation

Workplace

Selection of workplace using the workplaces dialog

Group from / to

Selection based on a group. Selection by drop-down list

Finished article

Restriction to a finished article

Order

Restriction to an order

OP

Restriction to a single operation

Automatic counter assignment (EMG 8.2)

Requirement: In the application "Assignment of counter to machine", the energy meters are assigned to workplaces. If this option is checked, you cannot select the energy meters manually. Following the assignment of counter to machine, the energy meters are identified and then displayed as result.

Detail application consumption correlation

The detail application includes two sections. The upper part shows the log records of operations, the lower section displays the consumption.

The log records of operations matching the selection criteria entered above (period from/to, article, order number/operation number) are displayed as individual bars in the detail application (1 row for each workplace/machine).

The user can select the displayed log records (multiple selection possible by using the Ctrl key). The resulting time slice/s is/are used for presenting the consumption.

The lower section of the detail application shows two sections with different information for each selected consumption meter.

Section 1:

This section shows the result of the correlation: The periods result from the selected order and multiple selection of order bars.

Section 2:

The consumption documents matching the selected consumption meter and period entered in the selection panel are shown. However, they are not displayed as complete document but rather as averaged point (average power in the document). This refers to the load profile of the consumption resource.

3 Status of Consumption Correlation

Overview

Menu	Resource management → Resource analysis → Status of consumption correlation
Transaction code	concorsta
Function authorization	concorsta

This document describes the application "Status of consumption correlation" in the Manufacturing Operation Center (MOC).

Application

The application "Status of consumption correlation" is new. The status of consumption correlation graphically shows the energy consumption in relation to the recorded statuses of workplaces/machines. You can present energy consumption in connection with the accrued information on statuses. This way, it is e.g. possible to identify the energy consumption during the setup or a disturbance.

Energy documents are assigned to the statuses by time and selection.

Please note:

The application only shows completed status documents.

Selection criteria

Reference

Selection of resources using the resource list

Date from / until

Selection of a period for consumption correlation

Workplace

Selection of workplace using the workplaces dialog

Group from / to

Selection based on a group. Selection by drop-down list

Report group

Selection based on a report group. Selection by drop-down list

Automatic counter assignment

If this option is checked, HYDRA identifies automatically the relevant energy counters following the counter to machine assignment.

Detail application status of consumption correlation

The detail application includes two sections. The upper part shows the status of machines/workplaces, the lower section displays the consumption.

The statuses matching the above mentioned selection criteria (period from/to, workplace, group/report group) are displayed as individual bars in the detail application (1 row for each workplace/machine).

The user can select the displayed status log records (multiple selection is possible by using the Ctrl key). The resulting time slice/s is/are used for presenting the consumption.

The lower section of the detail application shows two sections with different information for each selected consumption meter.

Section 1:

This section shows the result of the correlation: The periods result from the selected order and multiple selection of order bars.

Section 2:

The consumption documents matching the selected consumption meter and period entered in the selection panel are shown. However, they are not displayed as complete document but rather as averaged point (average power in the document). This refers to the load profile of the consumption resource.